Reverse triggering — Q21

Define reverse triggering and explain its mechanism

- Patient–ventilator dyssynchrony in which a passive, machine-initiated breath triggers an involuntary diaphragm contraction
- Mechanism: respiratory entrainment – a reflex coupling (stretch-reflex/vagal) between lung inflation and brainstem respiratory output, with a fixed phase relation (e.g., 1:1, 1:2)
- Typically seen in deeply sedated patients (e.g., ARDS) with no own triggering

How do you recognise it on the ventilator waveforms

- Volume control: concave dip in the pressure trace during or after inspiration of a mandatory breath
- If the effort persists into expiration → double triggering/breath stacking (VT up to double)
- Pressure control: increase in flow and volume during the breath
- Oesophageal pressure or diaphragm EMG confirms (effort follows the machine breath)

What are its consequences and how is it managed

- Breath stacking → high tidal volumes and transpulmonary pressures (VILI)
- Eccentric (lengthening) diaphragm contractions → diaphragm injury; pendelluft
- Management: adjust sedation (lighter or deeper), change RR or inspiratory time to break entrainment, switch to a patient-triggered mode
- Neuromuscular blockade if severe ARDS and unsafe volumes
- Can also be beneficial (prevents diaphragm atrophy) if VT controlled

Patient–ventilator dyssynchrony — Q22

Classify patient–ventilator dyssynchrony with examples

- Trigger: ineffective efforts (missed triggers), auto-triggering, reverse triggering, double triggering
- Flow: flow starvation (too little flow for demand) or excessive flow
- Cycling: premature cycling (machine ends breath early) or delayed cycling (breath continues into patient expiration)
- This patient has ineffective triggering

Explain the causes of ineffective triggering in COPD

- Intrinsic PEEP (auto-PEEP): patient must first generate pressure equal to PEEPi before the trigger is sensed
- Excess pressure support → large VT, long Ti → more air trapping and weaker drive
- Insensitive trigger setting
- Weak respiratory muscles, over-sedation, metabolic alkalosis

How would you correct it

- Apply external PEEP to ~80% of PEEPi to reduce trigger work
- Reduce pressure support to lower VT and allow full expiration
- Increase expiratory trigger (cycle-off) % (e.g., from 25% to 40–50%) to shorten Ti
- Use a more sensitive flow trigger; treat bronchospasm; consider NAVA/PAV

Transpulmonary pressure — Q23

Define transpulmonary pressure and give the formula

- Transpulmonary pressure (PL) = airway pressure − pleural pressure (estimated by oesophageal pressure, Pes)
- End-inspiratory PL = Pplat − Pes(insp)
- End-expiratory PL = PEEP(total) − Pes(exp)

Calculate her end-inspiratory and end-expiratory transpulmonary pressures and interpret them

- End-inspiratory PL = 34 − 20 = 14 cmH2O → safe (< 20–25 cmH2O); high Pplat is due to a stiff chest wall
- End-expiratory PL = 10 − 16 = −6 cmH2O → alveoli collapse at end-expiration (atelectrauma)
- She needs more PEEP (to make end-expiratory PL ≥ 0, i.e., PEEP ~16–18)
- Plateau pressure overestimates lung stress in obesity/abdominal hypertension

Explain the difference between lung stress and strain, and the role of the chest wall

- Stress = transpulmonary pressure (force per area on the lung)
- Strain = VT / FRC (change in volume relative to resting volume); harmful above ~1.5–2
- Airway pressure is shared between lung and chest wall according to their elastances: PL = Paw × EL / (EL + ECW)
- Stiff chest wall (obesity, abdominal hypertension, oedema) takes more of the pressure → lung stress lower than Pplat suggests

Transpulmonary pressure — Q24

Describe how an oesophageal balloon catheter is placed and its position checked

- Insert balloon catheter (nasal/oral) into the lower third of the oesophagus (~35–40 cm), inflate with the recommended small volume
- Positive pressure deflection when advanced into the stomach, then withdraw until cardiac oscillations appear
- Occlusion test (Baydur): during end-expiratory occlusion, ratio of ΔPes/ΔPaw with efforts (or chest compression if passive) should be 0.8–1.2
- Check the trace for cardiac artefact and correct balloon volume

Compare the two methods of using Pes to set PEEP and summarise EPVent-2

- Absolute (direct) method: set PEEP so that end-expiratory PL is 0 to +5 cmH2O
- Elastance-derived method: end-inspiratory PL = Pplat × (EL/ERS), keep < 20–25 cmH2O
- EPVent-2 (2019): Pes-guided PEEP vs high PEEP/FiO2 table – no difference in mortality or ventilator-free days (post-hoc benefit if PEEP near 0 PL)

What is driving pressure and why does it matter

- Driving pressure (ΔP) = Pplat − PEEP = VT / respiratory system compliance
- Reflects the tidal volume relative to the 'baby lung' size
- ΔP > 15 cmH2O strongly associated with mortality (Amato 2015); keep ≤ 14–15

ROX index — Q25

Calculate his ROX index and interpret it

- ROX = (SpO2 / FiO2) / RR = (92 / 0.6) / 34 = 153 / 34 ≈ 4.5
- ROX ≥ 4.88 at 2, 6 or 12 h predicts HFNC success (low risk of intubation)
- ROX < 2.85 at 2 h, < 3.47 at 6 h, < 3.85 at 12 h predicts failure
- 4.5 is between the cut-offs → reassess closely (repeat in 1–2 h); do not delay intubation if worsening

List the other signs of HFNC failure

- Rising RR, accessory muscle use, thoraco-abdominal paradox
- Falling ROX over time; rising FiO2 requirement
- Altered mental state, haemodynamic instability
- HACOR score > 5 at 1 h on NIV; rising lactate

Why is delayed intubation harmful in this patient

- Delayed intubation after HFNC failure is associated with higher mortality (Kang 2015)
- Strong inspiratory efforts cause patient self-inflicted lung injury (P-SILI) – large transpulmonary swings
- Intubation in extremis carries higher risk of hypoxaemia and arrest

High-flow nasal cannula — Q26

Explain the physiological effects of HFNC

- Flow up to 60 L/min matches patient inspiratory flow → stable, accurate FiO2 (up to 1.0)
- Washes out anatomical dead space → improved CO2 clearance and lower minute ventilation need
- Small flow-dependent PEEP (~0.5–1 cmH2O per 10 L/min with mouth closed) → alveolar recruitment
- Heated, humidified gas improves secretion clearance and comfort; reduces work of breathing

Summarise the evidence for HFNC

- FLORALI (2015): HFNC vs NIV vs standard O2 – no difference in intubation overall; lower 90-day mortality with HFNC; benefit in P/F ≤ 200
- Post-extubation: HFNC non-inferior to NIV in high-risk patients (Hernández 2016); combining NIV + HFNC lowers reintubation (HIGH-WEAN)
- Preoxygenation/apnoeic oxygenation during intubation

When would you choose NIV rather than HFNC

- Hypercapnic COPD exacerbation (pH < 7.35, PaCO2 > 45)
- Cardiogenic pulmonary oedema
- Immunocompromised – either HFNC or NIV acceptable
- Obesity hypoventilation; post-extubation in hypercapnic patients

RSBI and spontaneous breathing trial — Q27

Calculate his rapid shallow breathing index and interpret it

- RSBI = RR / VT (litres) = 32 / 0.25 = 128 breaths/min/L
- RSBI > 105 predicts weaning failure; < 105 predicts success (Yang & Tobin 1991)
- It is one tool alongside clinical assessment; SBT failure is likely here

List the criteria for readiness to wean and how an SBT is performed

- Cause of respiratory failure improving; PaO2/FiO2 > 150–200 on PEEP ≤ 8 and FiO2 ≤ 0.4–0.5
- Haemodynamically stable (no or low-dose vasopressor), awake/cooperative (RASS 0 to −1), adequate cough
- SBT: PS 5–8 cmH2O with PEEP ≤ 5, or T-piece, for 30–120 min
- Failure criteria: RR > 35, SpO2 < 90%, HR > 140, SBP change > 20%, agitation, sweating, accessory muscle use

Give four causes of weaning failure

- Respiratory load > capacity (bronchospasm, secretions, auto-PEEP)
- Cardiac: weaning-induced pulmonary oedema (raised BNP, echo)
- Neuromuscular: ICU-acquired weakness, diaphragm dysfunction, electrolytes (K, PO4, Mg)
- Neuropsychological: delirium, anxiety; metabolic alkalosis; over-feeding

RSBI and spontaneous breathing trial — Q28

List the risk factors for extubation failure

- Age > 65, chronic heart or lung disease
- Ventilation > 7 days, failed previous SBT
- Weak cough, copious secretions, poor neurological state
- Hypercapnia at end of SBT, BMI > 30, upper airway problems

How would you assess and manage the risk of post-extubation stridor

- Cuff leak test: leak < 110 mL (or < 12–24% of VT) suggests laryngeal oedema
- If no leak and high risk: methylprednisolone 20–40 mg IV 4-hourly (or similar), starting ≥ 4 h before extubation
- Post-extubation stridor: nebulised adrenaline, heliox, steroids; reintubate if needed

What support would you use after extubation and why

- High-risk patients (this patient): prophylactic NIV immediately after extubation, ideally alternating with HFNC
- Reduces reintubation rate (HIGH-WEAN 2019; ERS/ATS guideline)
- Low-risk patients: HFNC reduces reintubation vs conventional O2
- Do not use NIV as rescue for established post-extubation respiratory failure (delays reintubation, increased mortality)

Auto-PEEP — Q29

What is auto-PEEP and how do you detect it

- Positive alveolar pressure at end-expiration due to incomplete emptying (dynamic hyperinflation)
- Flow–time curve: expiratory flow does not return to zero before next breath
- Measure with an end-expiratory hold (pause): total PEEP − set PEEP = intrinsic PEEP (requires passive patient)
- In spontaneously breathing patients: oesophageal pressure drop before flow starts

Explain why her blood pressure has fallen

- High intrathoracic pressure reduces venous return to the right heart
- Raised pulmonary vascular resistance increases RV afterload
- Can cause PEA arrest; risk of barotrauma (tension pneumothorax)

Outline your immediate actions and ventilator strategy

- Disconnect from the ventilator briefly and allow full exhalation (± gentle chest compression); fluid bolus
- Reduce minute ventilation: RR 10–12, VT 6–8 mL/kg; accept permissive hypercapnia (pH > 7.2)
- Increase expiratory time: high inspiratory flow (60–100 L/min), short Ti, I:E ≥ 1:4
- Keep Pplat < 30, intrinsic PEEP < 10–15 cmH2O
- Treat bronchospasm: salbutamol, ipratropium, steroids, IV magnesium, ketamine; exclude pneumothorax

Auto-PEEP — Q30

Explain the 'waterfall' concept and how external PEEP helps him

- In COPD, expiratory flow limitation occurs because small airways collapse (critical closing pressure)
- Applying external PEEP below that pressure does not raise alveolar pressure (like water over a waterfall)
- External PEEP reduces the pressure gradient the patient must overcome to trigger → less work of breathing
- Set external PEEP ~80% of intrinsic PEEP (here ~8 cmH2O); check that plateau pressure does not rise

List the factors that increase dynamic hyperinflation

- High minute ventilation (high RR or VT)
- Short expiratory time (long Ti, low inspiratory flow)
- High airway resistance (bronchospasm, secretions, small ETT, kinked tube)
- Increased lung compliance (emphysema)

How does auto-PEEP affect measurements on the ventilator and at the bedside

- Overestimates CVP and PAOP (transmitted pressure)
- Underestimates respiratory compliance if PEEPi ignored (Crs = VT / (Pplat − total PEEP))
- Increases trigger work and causes ineffective triggering

Stress index — Q31

Define the stress index and explain how it is obtained

- Coefficient b of the power equation fitted to the airway pressure–time curve during constant-flow inspiration: Paw = a·t^b + c
- Requires volume control with constant flow, passive patient (no efforts)
- Reflects the change in compliance during tidal inflation

Interpret the three stress index values and state what you would do in each case

- b < 0.9 (upward concavity): compliance improves during inflation → tidal recruitment → increase PEEP
- b ≈ 0.9–1.1 (straight line): constant compliance → optimal
- b > 1.1 (downward concavity, curve bends upward): compliance falls → overdistension → reduce PEEP or VT
- Grothues/Ranieri: targeting SI 0.9–1.1 reduces inflammatory mediators

What are the limitations of the stress index

- Needs passive patient and square-wave flow
- Affected by chest wall stiffness (obesity, abdominal hypertension) – may reflect chest wall, not lung
- Global measure – does not detect regional heterogeneity
- Visual estimation is imprecise without software; endotracheal tube resistance affects proximal pressure

Stress index — Q32

Describe the Berlin definition of ARDS

- Onset within 1 week of a known insult or new/worsening respiratory symptoms
- Bilateral opacities not fully explained by effusion, collapse or nodules
- Not fully explained by cardiac failure or fluid overload
- Severity on PEEP ≥ 5: mild P/F 200–300, moderate 100–200, severe ≤ 100

Outline your lung-protective ventilation strategy

- VT 6 mL/kg predicted body weight (range 4–8)
- Plateau pressure ≤ 30 cmH2O; driving pressure ≤ 15 cmH2O
- PEEP set by ARDSNet higher PEEP/FiO2 table, or by stress index/compliance/Pes
- Permissive hypercapnia (pH > 7.2); target SpO2 88–95%

Describe the adjuncts for severe ARDS and the evidence for each

- Prone positioning ≥ 16 h/day when P/F < 150 (PROSEVA: mortality 16% vs 33%)
- Neuromuscular blockade for 48 h in early severe ARDS if dyssynchrony (ACURASYS benefit; ROSE no benefit)
- Conservative fluid strategy (FACTT)
- Avoid routine recruitment manoeuvres (ART trial: harm)
- VV ECMO for refractory hypoxaemia (EOLIA); steroids may be considered (dexamethasone in COVID; DEXA-ARDS)

Transpulmonary pressure — Q33

Explain how raised intra-abdominal pressure affects respiratory mechanics

- Diaphragm pushed cephalad → reduced FRC, basal atelectasis
- Chest wall elastance rises → higher Pplat for the same VT
- A large part of Pplat is used to distend the chest wall, so lung stress may be lower than Pplat suggests
- About 20–80% of IAP is transmitted to the thorax

How would you set PEEP and VT in this patient

- Measure Pes to calculate transpulmonary pressure (end-inspiratory < 20–25, end-expiratory ≥ 0)
- Higher PEEP may be needed to counter basal collapse
- Tolerate Pplat > 30 if transpulmonary pressure is safe

What measures would reduce intra-abdominal pressure

- Nasogastric and rectal decompression; prokinetics
- Sedation/analgesia ± neuromuscular blockade to relax abdominal wall
- Avoid fluid overload; diuresis or ultrafiltration
- Drain ascites/collections; surgical decompression if ACS

Prone positioning — Q34

Explain how prone positioning improves oxygenation and reduces lung injury

- More homogeneous distribution of transpulmonary pressure and ventilation (less dorsal collapse)
- Better V/Q matching: perfusion stays dorsal while ventilation moves dorsal
- Heart and abdomen no longer compress the dorsal lung
- Reduces overdistension and cyclic collapse (VILI); better secretion drainage

State the indications and contraindications

- Indication: moderate–severe ARDS, P/F < 150 on FiO2 ≥ 0.6 and PEEP ≥ 5, early (< 36 h), ≥ 16 h/day
- Absolute contraindication: unstable spinal fracture; open abdomen/chest relative; raised ICP
- Relative: haemodynamic instability, recent sternotomy, pregnancy, facial trauma, massive haemoptysis

List the complications of prone positioning

- Pressure sores (face, chest, knees), facial oedema
- ETT displacement/obstruction, line or drain dislodgement
- Haemodynamic instability on turning
- Brachial plexus injury, corneal abrasion, feed intolerance